Three-Dimensional Angiography Using Mobile C-Arm with IMU Sensor Attached: Initial Study

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***Abstract*—Three-dimensional (3D) computed tomography (CT) imaging is becoming an essential demand in several clinical procedures. Mobile C-arm is a useful imaging tool for image-guided interventional radiology. C-arm systems are provided with X-ray image intensifier (XRII) or flat-panel detectors. Essentially, C-arm CT systems requires scanners with flat-panel detectors for its ability to provide homogenous image quality and improve the resolution of low-contrast subjects compared to those equipped with XRII. However, C-arm systems with XRIIs are widely used in several interventional procedures. Such systems can provide a high quality two-dimensional (2D) fluoroscopic images that facilitates minimal invasive surgery. However, it is unable to provide depth information for 3D imaging due to several factors. First, the gantry of XRII-based C-arms is usually operated manually, where the rotation angle is determined using printed angle scale attached to the scanner gantry. Second, the gantry orbital rotation is normally limited to angular range less than theoretically required for exact 3D reconstruction. Third, considering the offset-scan geometry, which is common configuration in mobile C-arm with XRII, the number of rays passing through field-of-view (FOV) is limited. In this paper, we develop a 3D angiographic imaging system using commercial C-arm system equipped with XRII. First, an in-house made gantry rotation unit is developed to control the scanner orbital rotation. Second, the gantry rotation is traced using inertial measurement unit (IMU) sensor attached to the scanner gantry. Geometry information obtained from IMU sensor are used to define the gantry position in the 3D space and synchronized with detector measurements. The SCAN algorithm is used for the 3D reconstruction and achieved results are of high quality.**

I. Introduction

Three-dimensional (3D) computed tomography (CT) imaging is becoming an essential demand in several clinical procedures. Mobile C-arm is a useful imaging tool for image-guided interventional radiology. C-arm systems are provided with image intensifier (XRII) or flat-panel detectors. Essentially, C-arm CT systems requires scanners with flat-panel detectors for its ability to provide homogenous image quality and improve the resolution of low-contrast subjects compared to those equipped with image intensifier [1], [2]. However, C-arm systems equipped with XRIIs are widely used in several interventional procedures. Such systems can provide a high quality two-dimensional (2D) fluoroscopic images that facilitates minimal invasive surgery. However, it is unable to provide depth information for 3D imaging due to several factors. First, the gantry of XRII-based C-arms is usually operated manually, where the rotation angle is determined using printed angle scale attached to the scanner gantry. Second, the gantry orbital rotation is normally limited to angular range less than theoretically required for exact 3D reconstruction. Third, considering the offset-scan geometry, which is common configuration in mobile C-arm with XRII, the number of rays passing through field-of-view (FOV) is limited. Several methods are developed to solve image reconstruction challenges discussed in this paper (e.g. [3], [4])

Digital subtraction angiography (DSA) is an example of common application performed through X-ray C-arm system. In many cases, it is difficult to observe the exact blood vessels structures using 2D images acquired using DSA technique with XRII-based C-arm. In this paper, we develop a 3D angiographic imaging system using commercial C-arm system equipped with XRII. First, an in-house made gantry rotation unit is developed to control the scanner orbital rotation. Second, the gantry rotation is traced using inertial measurement unit (IMU) sensor attached to the scanner gantry. Geometry information obtained from IMU sensor are used to define the gantry position in the 3D space and synchronized with detector measurements. The SCAN algorithm [5] is used for the 3D reconstruction and achieved results are of high quality.

II. Materials and Methods

A. The C-arm system with XRII

We have implemented the proposed system using a clinical C-arm scanner (Ziehm Vision R, Ziehm Imaging GmbH, Nürnberg, Germany)[1]. The scanner is equipped with X-ray source and XRII with diameter of 31 cm. The gantry orbital rotation range is $115°$ and gantry angulation is $\pm 225°$. Both orbital and angulation movements are shown in Fig. 1. The result presented here is achieved from imaging session with orbital rotation with approximate angle range of $115°$ (Fig. 1(a)).

B. IMU sensor and gantry motion control unit

The x-IMU sensor consists of high-sensitivity gyroscope, accelerometer and magnetometer with data transfer available through wireless communication. In this study, we have used a commercial IMU sensor (x-IMU, x-io Technologies Ltd., UK)[2] shown in Fig. 2(a). An In-house software is developed for sensor data acquisition and synchronization with detector

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[1]http://www.ziehm.com/en/products/ziehm-vision-r/

[2]http://www.x-io.co.uk/products/x-imu/

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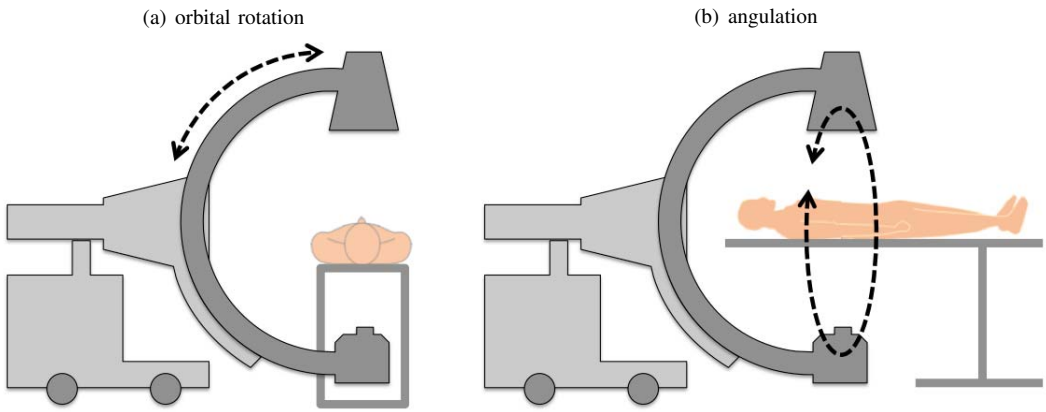

Fig. 1. Sketch of C-arm imaging setup and patient position with (a) gantry orbital rotation $-90°$:$+25°$ and (b) gantry angulation $\pm 225°$.

measurements using x-IMU API. The accelerometer and gyroscope data was used to define the 3D position of the scanner gantry in space.

The gantry is powered by two geared 35W DC motors each of 3.432 N.m torque through a system fabricated from high density polyethylene (HDPE). The rotational motion is transferred by the friction of 5 compressed rubber rings each of 70 mm diameter to avoid slipping as shown in Fig. 2(b). The motors speed and direction are controlled through an in-house designed USB controller either via foot pedals or a GUI. The orbital angle of the gantry is measured by OMRON E6B2-CWZ6C rotary encoder which leads to 3.33×10^{-3} degree precision. A PID controller is implemented in the GUI to set the desired orbital angle.

C. Image reconstruction algorithm

We have used the following imaging model

$$A\vec{x} = \vec{b}, \tag{1}$$

where $\vec{x} = (x_1, \ldots, x_n)$ is the image vector, $\vec{b} = (b_1, \ldots, b_m)$ is the projection data and $A = \{a_{ij}\}$ is an $m \times n$ system matrix modeling the scanner geometry. The sparsity-constrained angiography (SCAN) algorithm [5] is employed for image reconstruction. SCAN algorithm is developed using alternating directions method of multipliers (ADMM) to minimize the following cost function

$$\min_{\vec{x},\vec{z}} f(\vec{x}) + g(\vec{z}) \quad \text{subject to} \quad \vec{x} - \vec{z} = 0, \tag{2}$$

where $f(\vec{x})$ and $g(\vec{z})$ are defined by

$$f(\vec{x}) = \|\vec{x}\|_1^1 \quad \text{and} \quad g(\vec{z}) = \begin{cases} 0 & (\text{if } \vec{z} = \vec{b}) \\ \infty & (\text{otherwise}) \end{cases} \tag{3}$$

The SCAN algorithm [5] is given in the following form

where $\rho > 0$ is the weighting parameter of the second-order penalty term in augmented Lagrangian function corresponding to Eq.(2) and ART($\vec{t}$) is a single iteration of ART algorithm with initial image vector $\vec{t}$. The soft-thresholding function SoftThre(.) is defined as

$$[\text{SoftThre}(\vec{t}, a)]_j = \text{sign}(t_j)(|t_j| - a)_+ \tag{4}$$

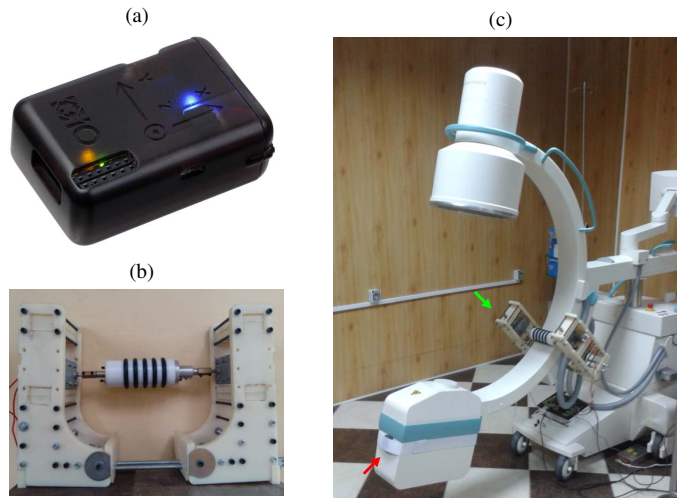

Fig. 2. (a) The x-IMU sensor with housing shield and battery, (b) the gantry motion control unit and (c) the 3D C-arm angiography system with x-IMU sensor attached (red arrow) and gantry motion control unit installed (green arrow).

1 Initialization: $\vec{x}^{(0)} = \vec{z}^{(0)} = \vec{\lambda}^{(0)} = (0, 0, \ldots, 0)^\top$
2 **for** $k = 0, 1, \ldots, K-1$ **do**
$\vec{x}^{(k+1)} = \text{SoftThre}(\vec{z}^{(k)} - \vec{\lambda}^{(k)}/\rho,\ \rho)$
$\vec{z}^{(k+1)} = \text{ART}(\vec{x}^{(k+1)} + \vec{\lambda}^{(k)}/\rho)$
$\vec{\lambda}^{(k+1)} = \vec{\lambda}^{(k)} + \rho(\vec{x}^{(k+1)} - \vec{z}^{(k+1)})$
end
3 Solution $\vec{z}^* = \vec{z}^{(K)}$

III. Experimental Results

We have implemented a real-data experiment to evaluate the proposed 3D angiographic system using clinical scanner installed in Suez Canal University Specialized Hospital. The x-IMU sensor is attached to the scanner x-ray tube using hook and loop strip and the automatic motion control unit is installed to the scanner gantry as shown in Fig. 2(c). A metal wire, shown in Fig. 3(a), is hanged on holder such that the spiral rotating head is located inside the FOV. The projection data is acquired using a step-and-shoot mode. The gantry is rotated to target angle then we acquire a single x-ray shot. We consider

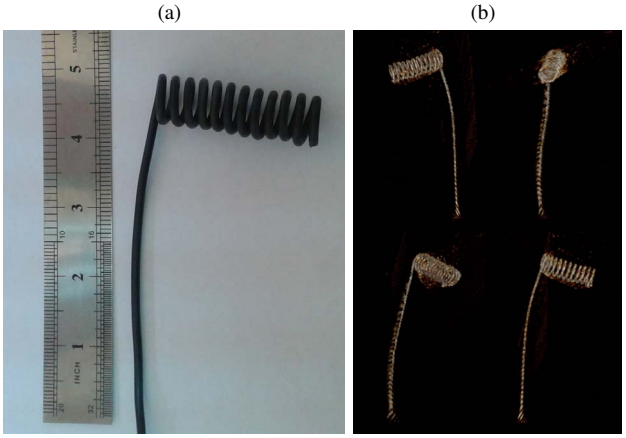

Fig. 3. (a) A photograph of the metal wire object used to simulate human artery enhanced with contrast agent (with scale meter). (b) Volume rendering of the 3D reconstructed object image displayed in different view positions.

angle shift of 5° over orbital rotation of 115° resulting in 24 projection views. The 3D geometric data representing the gantry position corresponding to each view angle is measured using the x-IMU sensor and transferred to a PC which is used for the 3D reconstruction. The SCAN algorithm was implemented with parameters $\rho = 20$ and $K = 10$. Volume rendering of the reconstructed 3D image with different view positions is shown in Fig. 3(b).

IV. Conclusion

In this study, we investigate the development of 3D angiography using mobile C-arm scanner with IMU sensor attached. The gantry motion is automatically controlled using in-house developed hardware. The SCAN algorithm is used for 3D image reconstruction from 24 projection views. Achieved results indicate a potential usefulness of the proposed system in clinical applications.

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